

Employee Management System

System Requirements Documentation

Ama Mensah-Dwumfua

Final Demo Submission

Table of Contents

- 1. Customer Problem Statement and System Requirements
- 2. Functional Requirements Specification
- 3. System Sequence Diagram
- 4. Activity Diagrams
- 5. User Interface Specification
- 6. Traceability Matrix
- 7. System Architecture and System Design
- 8. Algorithms and Data Structures
- 9. User Interface Design and Implementation
- 10. Design of Tests
- 11. Project Plan
- 12. References

1. Customer Problem Statement and System Requirements

Many small businesses need a simple way to organize employee records and track attendance. Manual tracking can lead to missing records, mistakes, and extra work for managers or HR staff. The Employee Management System solves this problem by allowing users to add employees, view employee records, record attendance, and view attendance history in a simple Java program.

Requirement	Priority	Description
REQ1	5	The system allows users to add employee records.
REQ2	5	The system allows users to view employee records.
REQ3	5	The system allows users to record attendance.
REQ4	4	The system allows users to view attendance records.
REQ5	4	The system saves data to text files.
REQ6	3	The system supports roles such as Manager, HR, and Employee.

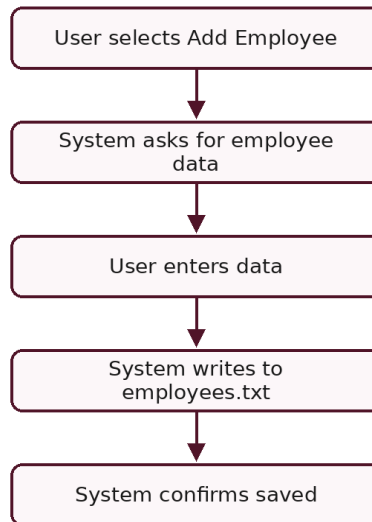
2. Functional Requirements Specification

- Add Employee: The user enters employee name, role, and department. The system saves this data in employees.txt.
- View Employees: The system reads employees.txt and displays all saved employees.
- Record Attendance: The user enters an employee name and chooses Clock In or Clock Out. The system saves this in attendance.txt.
- View Attendance: The system reads attendance.txt and displays saved attendance records.

3. System Sequence Diagram

The diagram below shows the sequence for adding an employee. The user selects the action, enters employee information, and the system writes the record to the text file automatically.

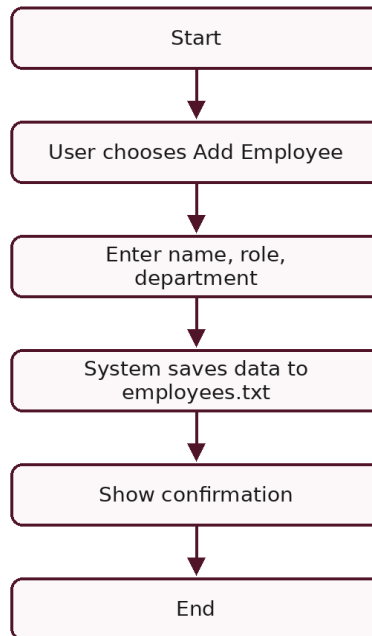
System Sequence: Add Employee



4. Activity Diagram - Add Employee

This activity diagram shows how the system adds an employee record from the menu and stores it in the employee text file.

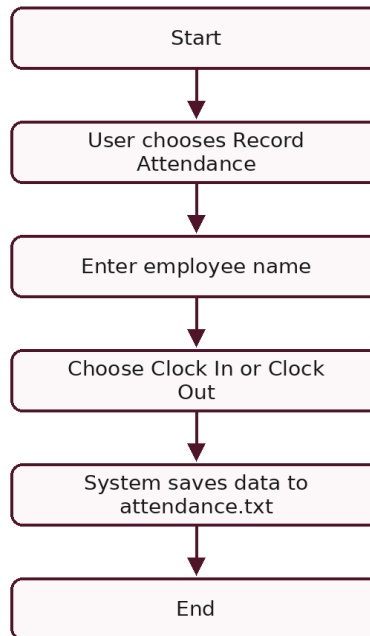
Activity Diagram: Add Employee



4. Activity Diagram - Record Attendance

This activity diagram shows how the system records attendance and stores the attendance entry in the attendance text file.

Activity Diagram: Record Attendance



5. User Interface Specification

The EMS uses a simple console interface. The user selects an option from a numbered menu. This keeps the system easy to use and easy to demonstrate.

Use Case	Navigation	User Effort
UC1 Add Employee	Menu -> Add Employee -> enter details	1 menu choice + typing
UC2 View Employees	Menu -> View Employees	1 menu choice
UC3 Record Attendance	Menu -> Record Attendance -> choose status	1 menu choice + typing
UC4 View Attendance	Menu -> View Attendance	1 menu choice

6. Traceability Matrix

Requirement	UC1	UC2	UC3	UC4
REQ1 Add Employee	X			
REQ2 View Employees		X		
REQ3 Record Attendance			X	
REQ4 View Attendance				X
REQ5 Save Data	X	X	X	X
REQ6 Roles	X	X		

7. System Architecture and System Design

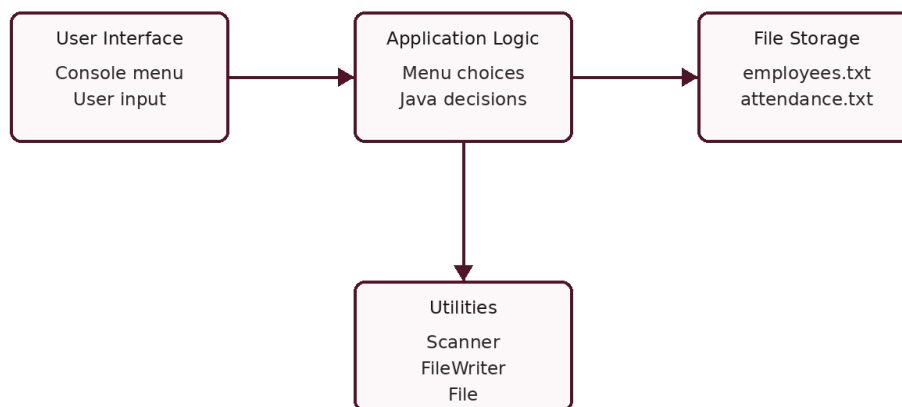
The EMS uses a simple layered style. The user interface is the console menu. The application logic is the Java code that checks the user's choice and performs the action. The storage layer is made of text files.

EMS System Block Diagram



UML Package Diagram

UML Package Diagram



The system runs on one computer. It does not use a network protocol. The persistent objects are employee records and attendance records. The storage strategy is flat text files.

8. Algorithms and Data Structures

The system uses a menu loop that continues until the user chooses Exit. It uses Scanner to read keyboard input and to read file content. It uses FileWriter to save new records. The main data structures are strings and text files.

9. User Interface Design and Implementation

The user interface is a simple numbered console menu. This design was selected because it is easy to understand and reduces unnecessary clicks. The system gives clear confirmation messages after saving information.

```
Employee Management System
1. Add Employee
2. View Employees
3. Record Attendance
4. View Attendance
5. Exit

Enter choice: 1
Enter employee name: Annora Buabeng
Enter role: Manager
Enter department: HR
Employee saved.
```

10. Design of Tests

Testing was completed manually by running each feature from the menu. Each feature was tested with sample data. The main test was to make sure the system saves data and then retrieves it correctly from the text files.

- Add Employee was tested by adding a new employee and checking employees.txt.
- View Employees was tested by reading the saved employee list.
- Record Attendance was tested by saving Clock In and Clock Out records.
- View Attendance was tested by displaying saved attendance records.

11. Project Plan

The project plan focused on building the system in small steps. First, the main menu was created. Next, employee storage was added. Then attendance storage was added. After that, documentation, testing notes, brochure, and presentation slides were completed.

Phase	Task	Status
1	Create project idea and requirements	Complete
2	Create diagrams and UI specification	Complete
3	Build Java console program	Complete
4	Add text-file storage	Complete
5	Prepare final demo materials	Complete

12. References

- Oracle. Java Platform Documentation. <https://docs.oracle.com/en/java/>
- Java File I/O concepts from course materials and Java documentation.
- System architecture concepts from course materials and the assigned architecture reading.